A model-agnostic dual-path supervisory logic for transformer differential protection, evaluated with a leak-free benchmark of six reproduced methods

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Abstract

Learning-based transformer differential relays are rarely compared under the same data, folds and decision logic. We evaluate on simulated 300 MVA and 40 MVA plants with trace-grouped cross-validation (3 seeds × 5 folds, 3,302 traces) and re-implement six published methods at their own sampling rates. Natively, none reaches 93 % dependability and 98.8 % security together: the two most dependable reach 97.0 % and 96.5 % at 75.3 % and 82.9 % security. We then study a dual-path supervisory logic (an 87T vote gating a confirmed classifier output, a classifier-alone path, a high-set element) with interchangeable classifiers. With an equal search and epoch budget, six classifier instances reach 93.5–95.7 % dependability and 98.8–99.9 % security. All six are more dependable than a validation-tuned harmonic-restrained 87T (91.2 % / 99.1 %) by 2.3–4.6 points (post-hoc, trace-clustered test, Holm-adjusted), with security within −0.3 to +0.8 points. The deployed LSTM instance of our Simulink closed loop (93.0 % / 99.0 %) is 1.8 points more dependable than the tuned relay ($p_{\mathrm{H}} = 0.005$), but its lead is not stable across seeds (−0.2 to +2.8 points). Transfer to the unseen plant depends on the classifier: zero-shot security is 99.9 % with Key's network but 88.4 % with the deployed LSTM, whose classifier-alone path caused 87 of 89 false trips. High-impedance internal faults remain a limit (dependability about 65 % deployed, 62 % tuned relay). The supervisory logic and the evaluation protocol, not the choice of classifier, are the contribution.

Keywords: transformer differential protection, inrush, CT saturation, machine learning, supervisory logic, benchmark, grouped cross-validation

1. Introduction

Percentage-restrained differential protection (87T) is the main protection of power transformers. Magnetising inrush, over-excitation and current-transformer (CT) saturation create differential current without an internal fault, so practical relays add harmonic restraint or blocking, and these measures trade security against speed and against sensitivity to faults with low current [1]. Many studies replace or supplement the harmonic criteria with a learned classifier on wavelet or raw-current features [2, 3, 4, 5, 6, 7, 8, 9].

Two things limit what such studies show. First, they are rarely compared with each other under the same data, folds and decision logic, and the split unit (case or window) is often not stated; windows of one simulated case in both training and test sets inflate accuracy [10, 11]. Second, the learned model usually *is* the relay. When it errs, nothing in the protection logic bounds the error, and transfer to a transformer that was not in the training data is seldom tested.

This paper treats both problems together. The relay logic is a small supervisory layer that any classifier can plug into: an 87T vote gates a confirmed classifier output, a separate high-confidence path lets the classifier trip alone, and a high-set element trips without it. The evaluation is a benchmark in which six published methods are re-implemented on the same simulated data and scored with the same trace-grouped folds as the proposed logic and as a validation-tuned harmonic-restrained 87T. The contributions are:

1. a *leak-free benchmark* for learned 87T schemes: trace-grouped cross-validation (CV) over 3 seeds, validation-only thresholds, a shifted-range test set scored once, zero-shot transfer to a second transformer, six reproduced methods in their native and in a like-for-like mode, and paired tests whose unit is the trace (Sections 3–5);
2. a *dual-path supervisory logic* evaluated with 7 interchangeable classifiers, which shows that its operating point depends little on the classifier, while its transfer to another plant depends strongly on it (Sections 4 and 6);
3. an analysis of *where the supervision helps*: the 87T gate adds security for a poorly calibrated classifier and costs dependability for a well-calibrated one, and the classifier-alone path is the weak point under transfer (Section 7).

Section 2 places the logic among prior hybrid schemes, Section 3 describes

the plants and data, Section 4 the logic, Section 5 the protocol, Section 6 the results, Section 7 their interpretation and Section 8 the limitations.

2. Related work

How a learned model meets the 87T. The searched literature shows four patterns. (i) The model *restrains or blocks* the 87T: Rebizant et al. combine neural inrush detectors with second-harmonic restraint in one stabilising unit [12], and fuzzy and network-based stabilisation follow the same idea [13]. (ii) The model *conditions the input* of the 87T: Lin et al. mark the saturated samples of the differential current with an attention network and estimate the fundamental on the rest, so that a ratio-differential element decides without an inrush-blocking module [14]. (iii) The model *adapts the settings*: Rezaei recognises sub-synchronous resonance and ferroresonance and changes the relay settings for them [15]. (iv) The model *is* the relay, as in most recent work [3, 16, 6, 7, 17, 8, 9, 18], sometimes with a confirmation counter on its output [19] or an ANN that issues the trip next to the differential relay [20]. In line protection the mirror image appears: a learned model vetoes a conventional trip [21] or supervises a zone [22]. To our knowledge, prior work uses the learned model to restrain the differential element, to condition its input, to adapt its settings, or to replace it; here the 87T vote gates the learned decision, with a separate high-confidence path and a persistence filter on the output. The parts are known; the combination and its evaluation are what this paper adds.

Evaluation practice. Grouped (blocked) cross-validation is the standard remedy against dependent samples [23], and leakage between training and test data is a documented source of overstated machine-learning results [10, 11]. Paired classifier comparisons call for McNemar-type tests [24, 25, 26] with a correction for multiple comparisons [27]. Generalisation across systems is examined by knowledge-based networks [28] and, for line protection, by an open benchmark with committed partitions [29]. In the transformer-protection papers above, significance tests and cross-transformer transfer tests are rare, and baselines are re-implemented without a stated training budget.

Methods not reproduced. The benchmark reproduces six methods for which a paper gives a complete specification (Section 5). The MRA–BiGRU and VMD–CNN–BiLSTM schemes [7, 17] belong to the wavelet plus recurrent family represented by the Alhamd reproduction; the fast GRNN

[16] differs from the reproduced CNN of the same group [3] in the cell and the loss, and a GRU with the same budget is part of the same-feature study; the signal-localised CNN [6] is a CNN on wavelet coefficients, the family of the 1D-CNN instance; variable-window reinforcement learning [18] and methods that need the magnetising-branch voltage [28] are outside a current-only relay. Deterministic wavelet 87T schemes [30, 31] were not implemented.

3. Plants and data

Plants. Plant A is a 300 MVA, 230/11 kV Yd1 transformer, plant B a 40 MVA, 132/33 kV Dyn11 transformer, both Simulink models with a saturable transformer core, six saturable CTs, a source impedance, a load, breaker-controlled energisation and fault blocks on both sides. The leakage reactance is 11 % (A) and 10 % (B) with $X/R \approx 20$. Before any data were generated, each plant was checked against 6 physics criteria (true through-fault current, CT linearity at nominal burden, CT saturation with DC offset and with increased burden, voltage under load, CT ratio drift). The true steady through-fault current is 7.39 pu (A) and 8.64 pu (B). Plant A meets all 6 criteria; plant B fails 2 of 6 (a fully offset fault does not saturate its CTs at nominal burden, and its ratio drift at three times the burden exceeds the limit), and is reported as measured.

Scenarios. Each trace is 1.1 s at 1600 Hz. 16 categories cover normal load, energisation inrush (no-load, loaded, recovery after an external fault), over-excitation, external faults (solid, high-impedance, with CT saturation, during inrush) and internal faults (low- and high-voltage side, solid, high-impedance, evolving, switch-on-to-fault, with CT saturation, during inrush). Load, power factor, noise, CT mismatch and burden, source strength, frequency, inception angle, residual flux and fault resistance are drawn per trace. The "high-impedance" internal faults are terminal faults through a fault resistance drawn from bins of a sensitivity sweep; they are not turn-to-turn faults. Sympathetic inrush between parallel transformers and CT remanence are not modelled.

Sets. Plant A has 3,332 distinct traces after deduplication; the reproductions need a 20 kHz re-simulation, which failed for 30, so every comparison uses the common set of 3,302 traces (1,239 detectable faults, 1,970 healthy traces). A shifted-range set of 299 plant-A traces (extreme load, weak source, frequency offset, extreme burden, severe over-excitation, high noise) is scored

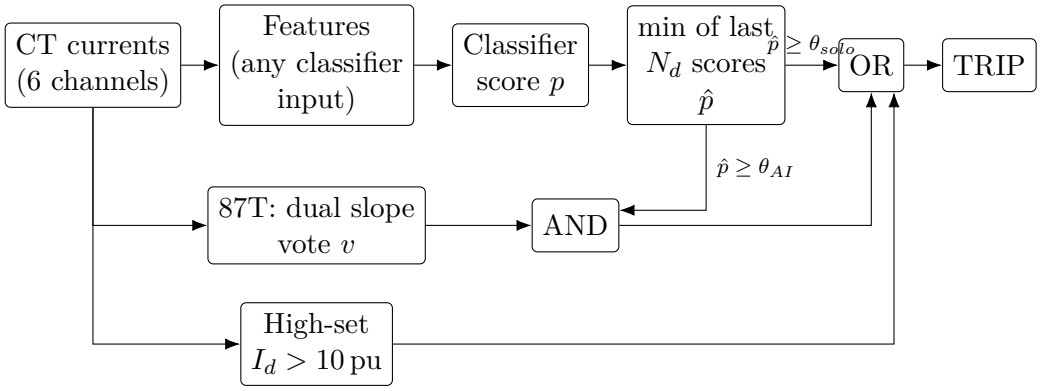


Figure 1: Dual-path supervisory logic. Any classifier that returns a fault score at the relay's decision instants can be used; $\hat{p}$ is the minimum of its last N_d scores.

once by each final model. Plant B provides 1,257 evaluable traces for zero-shot transfer (32 undetectable faults excluded).

4. Dual-path supervisory logic

The relay (Fig. 1) trips when

$$\mathrm{TRIP} = \mathrm{HighSet} \ \vee \ \Big(v \wedge \hat{p} \geq \theta_{AI}\Big) \ \vee \ \Big(\hat{p} \geq \theta_{solo}\Big), \tag{1}$$

where v is the any-phase vote of a dual-slope 87T (pickup 0.30 pu, slopes 25 % and 60 %, breakpoint 3 pu), $\hat{p}$ is the minimum of the last $N_d = 20$ classifier scores (12.5 ms at 1600 Hz), and the high-set element trips above 10 pu without the classifier. The gated path lets the classifier veto 87T operations on inrush, over-excitation and CT saturation; the classifier-alone path exists because a fault below the 87T pickup produces no vote. The same per-unit settings are used on both plants.

Thresholds. For every model, θ_{AI} and θ_{solo} are chosen on its validation traces by a plateau rule: among grid thresholds whose validation dependability and security are within 2 and 1 points of their maxima, the rule takes the longest contiguous run and returns its centre in logit space. If no threshold on the solo grid meets the rule, the classifier-alone path stays off. In the final plant-A model of the deployed instance $\theta_{AI} = 0.9950$ and $\theta_{solo} = 0.9980$.

Deployed instance. The instance used in the Simulink closed loop is an LSTM [32] with 2 layers of 64 units, layer normalisation, additive attention over 32 time steps and a two-layer output head (58,241 parameters). Its

input is 32 steps of 12 features of the plant-compensated per-unit differential current: per phase, the energies of a two-level Daubechies-4 decomposition [33] of a 20 ms window and the second-harmonic ratio. Its architecture was chosen in an earlier study on older data, outside the protocol below; the other instances were configured inside it.

5. Benchmark protocol

Splits and labels. Folds are stratified by scenario and contain whole traces, never windows; 5 folds are repeated with 3 seeds (15 fold-models per method), and every method uses identical folds. Within each training set, 15 % of the traces are held out; they serve both early stopping and threshold tuning, a mild optimism on validation shared by every learned model. The fault label starts at the applied fault time, verified on the waveform by the generator check; healthy traces are negative throughout. A trip before the fault counts as a miss. Faults whose compensated differential current rises by less than 0.05 pu are reported separately and excluded from dependability. Dependability is the share of detectable fault traces tripped, security the share of healthy traces not tripped.

Reproduced methods. Six methods were re-implemented from written specifications and trained on our data at each paper's sampling rate: Simões et al. [4], Ngaopitakkul and Kunakorn [34], Abdusalam et al. [5], Afrasiabi et al. [3] with the CT compensation stage of Hajipour et al. [35], Key et al. [9] and Alhamd et al. [8]. Each is scored in its *native* mode (the paper's own decision rule) and in a *like-for-like* mode (the same classifier with the relay's dwell, a validation-tuned threshold and the high-set element). Details that the papers leave open were fixed in writing before the code was run; specifications, amendments and sanity checks are in Supplementary Section S2.

Same-feature classifiers. SVM, random forest, MLP, 1D-CNN, GRU and LSTM on the same 12 features are evaluated standalone and inside the logic, once with default configurations (B1) and once tuned with an equal budget (B2): 12 configurations drawn per model (10–11 unique; LSTM 10), at most 25 epochs with patience 5 (about 10 used). The B2 search used validation traces of seed 1, folds 1–3, which are test traces of other fold-models, so it is not nested; SVM and random forest were fitted on 30,000 and 50,000 subsampled windows; B2 used window stride 4, the deployed LSTM stride 8. The budget is therefore equal in configurations and epochs, not in data, and the like-for-like reference for B2 is the tuned LSTM.

Conventional relays. The unrestrained dual-slope 87T; a per-phase harmonic-restrained 87T with textbook settings (pickup 0.30 pu, second harmonic 15 %, fifth harmonic 35 %); and the same relay tuned per fold on validation traces over a grid of pickup, k_2 and k_5 with the plateau rule.

Shifted range and transfer. Each final model is trained on all plant-A main traces except its validation share and scores the shifted-range set once. On plant B, the plant-A model is applied with its thresholds frozen (zero-shot); plant-B outcomes are never used to choose a setting.

Statistics. Because the 3 CV repetitions reuse the same traces, a pooled McNemar test would count each trace up to 3 times. All reported significance therefore comes from a trace-clustered sign-flip permutation test: the per-trace difference between two methods (correct minus correct) is averaged over the repetitions, and its mean is compared with 200,000 random sign flips [25, 26]. The test unit is the trace. p-values are Holm-adjusted [27] within the families stated in each table note (p_{H}). Decision rules were fixed in a dated plan before each analysis. Analyses that were added after earlier results had been seen (the instance-versus-tuned-harmonic family, the comparisons with the deployed instance in B2, the gated-only ablation and the latency re-measurement of one reproduction) are labelled post-hoc wherever they are reported.

AI-assisted code and analysis. The evaluation code, the reproduction modules, the statistics scripts, the table and figure generators and an independent audit of the results were written with an AI coding assistant ([**AUTHOR:** tool name, version and developer]). The assistant worked on the repository under written project rules: numbers enter the manuscript only through scripts that read saved result files; protocol and thresholds may not be changed after results are seen; every literature reference must resolve to a publisher record. The author specified the experiments, reviewed the code, the generated tables and the dated plan, and ran the simulations and trainings. All prompts, plans and logs are kept in the repository ([**AUTHOR:** confirm what is published]).

6. Results

6.1. Six reproduced methods

In their native modes the six methods span a wide range of operating points (Table 1, Fig. 2). The most dependable, Key et al. (97.0 % / 75.3 %) and Alhamd et al. (96.5 % / 82.9 %), trip on many healthy traces, and the

Table 1: Six published methods re-implemented and trained on the benchmark data (native decision logic and like-for-like mode), against the proposed relay. Dependability / security in %; CV = plant A, 3 seeds × 5 folds, common set (3,302 traces), mean over seeds; OOD and plant B: one final model scored once (plant B zero-shot, logic and threshold frozen from plant A). The reproductions are re-implementations trained on our data, not the authors' code.

Method	Mode	CV		OOD		Plant B		Rate	Latency per decision (median / budget)
		dep.	sec.	dep.	sec.	dep.	sec.		
Simões et al. 2021 (RT-BSWT + 2 SVMs; C, γ searched)	native	69.7	92.2	70.9	73.8	83.2	89.4	12.8 kHz	7.5 / 78.1 µs per sample; SVM 1 498 µs per detection
	like-for-like	76.8	72.1	75.0	60.4	86.7	71.7		
Ngaopitakkul & Kunakorn 2006 (db4 DWT + BPNN)$^{\triangle}$	native‡	27.7	99.3	20.3	99.3	28.0	97.6	20 kHz	3.3 / 50 µs per sample
	like-for-like	38.4	97.5	35.1	98.0	61.3	90.5		
Abdusalam et al. 2022 (DWT + OMP + BA + SVM)	native	76.4	99.9	83.1	94.0	71.4	99.7	10 kHz	609 / 2.5 ms per window: not real-time
	like-for-like	79.2	92.9	84.5	86.6	89.2	95.5		
Afrasiabi et al. 2020 (CNN, with Hajipour CT stage)	native‡	86.7	88.6	95.3	94.6	87.5	75.2	5.6 kHz	117 / 1250 µs per decision
	like-for-like	87.3	99.6	86.5	100.0	93.9	99.9		
Key et al. 2025 (Com-DNN + Dis-DNN)	native‡	97.0	75.3	96.6	78.5	98.2	69.7	3.2 kHz	37.0 / 312.5 µs per sample
	like-for-like	97.2	99.6	98.0	96.0	96.9	98.9		
Alhamd et al. 2024 (RT-BSWT + BiLSTM)	native‡	96.5	82.9	97.3	87.2	92.6	83.8	3 kHz	1166–1169 / 1667 µs per decision (trained configs; largest search-space config 1799 µs)
	like-for-like	97.9	98.2	95.3	95.3	90.0	96.2		
Proposed relay (deployed LSTM, dual-path)	hybrid	93.0	99.0	98.0	96.0	92.7	88.4	1.6 kHz	140 / 625 µs (ONNX)
LSTM alone (same logic, no 87T path)	standalone	97.7	95.6	99.3	87.3	92.9	87.3		

Notes. Native = the method's own decision rule; like-for-like = the same classifier with the relay's 20-sample dwell, validation-tuned threshold and high-set element. CV on the 3,302-trace common set (traces with a 20 kHz re-simulation). OOD and plant B scored once with frozen thresholds. ‡ fold-model s.d. > 3 pt. $^{\triangle}$ detector sanity check fails with measurement noise. Latency: single-thread Python on a development PC. Paired tests in Supplementary Table S3 (Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 48 tests.) Full notes: Supplementary Table S1.

most secure, Abdusalam et al. (76.4 % / 99.9 %), misses many faults. 0 of the six reaches 93 % dependability and 98.8 % security together (the joint criterion of the dated plan). The like-for-like mode changes this: Key's classifier with the relay's dwell, threshold and high-set reaches 97.2 % / 99.6 %, and Alhamd's 97.9 % / 98.2 %; 1 reproduction (Key et al.) meets the joint criterion in this mode. Much of the gap between published methods and a protection relay is therefore in the decision rule, not in the classifier. Two reproductions are tested off their design point: the lumped plants carry little energy above 1.6 kHz, which limits the high-frequency detectors of Simões et al. and Ngaopitakkul and Kunakorn, and the latter's detector fails its own sanity check once measurement noise is present.

6.2. Conventional relays

Table 2: Conventional relays on the benchmark data (dependability / security, %): unrestrained dual-slope 87T, harmonic-restrained 87T with textbook settings, and harmonic-restrained 87T tuned on validation traces, against the proposed relay. Plant-A CV (3 seeds × 5 folds, common set), OOD (final scoring) and plant-B zero-shot.

Relay	CV		OOD		Plant B	
	dep.	sec.	dep.	sec.	dep.	sec.
87T unrestrained (dual slope, 0.30 pu pickup)	80.3	63.5	89.3	53.3	89.5	62.0
Harmonic 87T, textbook (0.30 pu, k_2 0.15, k_5 0.35)	86.0	99.4	94.0	88.7	94.1	99.0
Harmonic 87T, tuned on validation	91.2	99.1	n/e	n/e	n/e	n/e
Proposed relay (deployed LSTM hybrid)	93.0	99.0	98.0	96.0	92.7	88.4

Notes. Grid boundary: pickup at the lowest grid value in 15 of 15, k_2 at the highest in 9 of 15, k_5 at the lowest grid value in 15 of 15 fold-models, so a wider grid might raise the tuned relay's dependability (not tested). Tuned minus proposed: dependability -1.78 pt ($p_{\mathrm{H}} = 5 \times 10^{-3}$), security +0.12 pt ($p_{\mathrm{H}} = 0.68$); textbook minus proposed: -6.99 pt ($p_{\mathrm{H}} \leq 2 \times 10^{-5}$) / +0.44 pt ($p_{\mathrm{H}} = 0.11$). Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 4 tests. n/e = not evaluated. Full notes: Supplementary Table S1.

Tuning the harmonic 87T on validation traces raises its dependability from 86.0 % to 91.2 % at 99.1 % security (Table 2). The deployed LSTM instance is 1.78 points more dependable ($p_{\mathrm{H}} = 5 \times 10^{-3}$; the per-seed lead ranges from -0.16 to +2.82 points) and not different in security ($p_{\mathrm{H}} = 0.68$). Against the textbook settings its lead is 7.0 points. The tuned settings sit

at the edge of the grid in every fold-model, so a wider grid might narrow the gap further.

6.3. Instances of the logic

Table 3: Instances of the dual-path supervisory logic with interchangeable classifiers (dependability / security, %): plant-A CV (3 seeds × 5 folds, common set, mean over seeds), OOD and plant-B zero-shot where a final model was scored; difference to the validation-tuned harmonic 87T (91.2 / 99.1; post-hoc, Holm over 16 tests); and the gated-only ablation (AI-alone path off).

Instance (hybrid mode)	CV		Δ dep. vs tuned	Δ sec. vs tuned	OOD		Plant B	
	dep.	sec.	harmonic 87T (pt)	harmonic 87T (pt)	dep.	sec.	dep.	sec.
LSTM, deployed (closed-loop configuration)	93.0	99.0	n/e	n/e	98.0	96.0	92.7	88.4
LSTM, tuned (B2)	93.6	99.1	+2.42*	+0.03†	n/e	n/e	n/e	n/e
1D-CNN, tuned (B2)	95.7	99.0	+4.57*	-0.05†	n/e	n/e	n/e	n/e
GRU, tuned (B2)	95.6	98.8	+4.41*	-0.25†	n/e	n/e	n/e	n/e
MLP, tuned (B2)	94.9	99.2	+3.74*	+0.14†	n/e	n/e	n/e	n/e
Random forest, tuned (B2)	93.5	99.5	+2.31*	+0.42	n/e	n/e	n/e	n/e
Key et al. network (Dis-DNN output)	94.1	99.9	+2.93*	+0.80*	98.0	95.3	95.5	99.9
Ablation, AI-alone path off (gated only; post-hoc, from saved decisions)								
LSTM, deployed, gated only	88.4	99.4			94.6	98.7	90.5	99.2
Key et al. network, gated only	n/e	n/e			94.6	95.3	94.1	100.0

Notes. * $p_{\mathrm{H}} < 0.05$ (post-hoc family of 16 tests: 12 added after the tuned-harmonic comparison plus its 4); † per-seed differences change sign. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 16 tests. 6 of 6 instances are significantly more dependable than the tuned harmonic 87T (+2.3 to +4.6 pt, $p_{\mathrm{H}} \leq 1 \times 10^{-3}$); Key's network significantly more secure; none is significantly less secure. In 3 of 15 CV fold-models of the deployed LSTM the plateau rule switched the AI-alone path off (AI threshold 0.75–0.999), so its CV row mixes dual-path and gated-only fold-models. Plant B false trips of the deployed hybrid: 87 of 89 via the AI-alone path. Gated only = AI-alone path off (post-hoc). n/e = not evaluated. Full notes: Supplementary Table S1.

Inside the same logic, the 6 instances reach 93.5–95.7 % dependability and 98.8–99.9 % security (Table 3). All of them are significantly more dependable than the tuned harmonic 87T, by 2.3–4.6 points ($p_{\mathrm{H}} \leq 1 \times 10^{-3}$, post-hoc), with security differences from -0.3 to +0.8 points; only Key's network is significantly more secure, and none is significantly less secure. The deployed LSTM is the least dependable of the good instances; it is kept because it is the instance of the Simulink closed loop. Only the kernel SVM (87.0 % / 97.7 %) falls clearly outside the cluster.

6.4. Same features, different classifiers

With identical features and an equal search budget (Table 4), the tuned CNN, GRU and MLP hybrids are more dependable than the tuned LSTM hybrid (+2.2, +2.0 and +1.3 points; $p_{\mathrm{H}} \leq 1 \times 10^{-4}$, $p_{\mathrm{H}} \leq 1 \times 10^{-4}$, $p_{\mathrm{H}} = 0.02$),

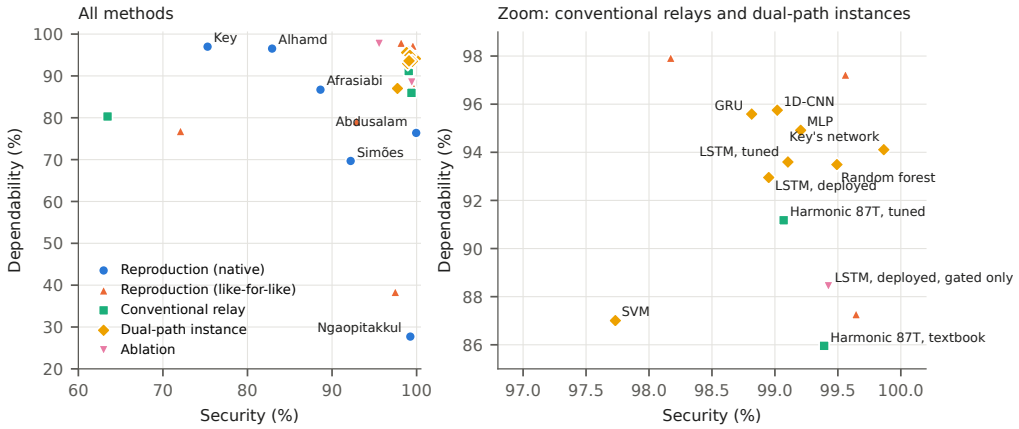


Figure 2: Dependability versus security on the plant-A CV (common set) for all reproductions (native and like-for-like), conventional relays, instances of the dual-path logic and the ablations. Right: zoom on the region of the conventional relays and the instances.

Table 4: Same 12 DWT + second-harmonic features, different classifiers: default configuration (B1: equal training routine) and tuned with an equal 12-configuration search (B2); standalone and inside the dual-path logic (hybrid). Dependability / security in %, plant-A CV, 3 seeds × 5 folds, common set, mean over seeds.

Classifier (architecture)	Default configuration				Tuned, equal budget				unique configs of 12	chosen configuration
	standalone		hybrid		standalone		hybrid			
	dep.	sec.	dep.	sec.	dep.	sec.	dep.	sec.		
SVM (pooled features)	84.9	75.1	81.7	96.9	90.9	90.4	87.0	97.7	10	n components 240, gamma x 1.0, alpha 1e-05
Random forest (pooled features)	96.8	95.7	92.4	99.1	97.1	95.4	93.5	99.5	11	n estimators 400, max depth None, min samples leaf 1
MLP (pooled features)	97.2	96.5	93.1	99.1	97.7	97.1	94.9	99.2	11	hidden1 64, hidden2 64, dropout 0.1, lr 0.001
1D-CNN (sequence)	97.1	97.0	93.7	99.2	98.0	97.3	95.7	99.0	10	channels 64, kernel 5, layers 3, dropout 0.1, lr 0.001
GRU + attention (sequence)	97.6	95.6	93.9	98.8	98.0	97.3	95.6	98.8	11	hidden 128, layers 1, dropout 0.1, lr 0.001
LSTM + attention (sequence)	97.7	95.6	93.0	99.0	98.4	97.3	93.6	99.1	10	hidden 128, layers 2, dropout 0.2, lr 0.003

Notes. Default = B1 run (stride 8; LSTM = deployed model). Tuned = B2: 12 drawn configurations per model, chosen on validation traces of seed 1, folds 1–3 (not nested), 25 epochs / patience 5, stride 4; SVM / RF fitted on 30,000 / 50,000 subsampled windows. Tuned hybrid minus tuned LSTM hybrid (dep. / sec., pt): CNN +2.2 / -0.1 (n.s.); GRU +2.0 / -0.3 (n.s.); ANN +1.3 / +0.1 (n.s.); RF -0.1 (n.s.) / +0.4 (n.s.); SVM -6.6 / -1.4. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 22 tests. Full notes: Supplementary Table S1.

and none differs significantly in security. Tuning the LSTM does not change it (+0.65 points, $p_{\mathrm{H}} = 0.72$). The ranking of classifiers inside the logic

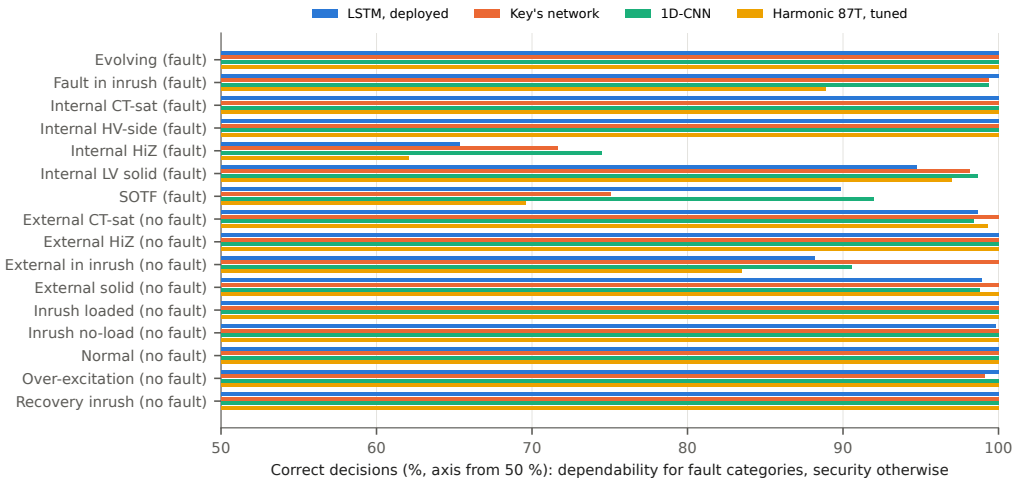


Figure 3: Correct decisions per scenario category (plant-A CV, common set, pooled over seeds): dependability for fault categories, security for the others. The axis starts at 50 %.

is therefore real but small next to the change from standalone to hybrid mode, which costs every classifier several points of dependability and raises its security.

6.5. Categories and high-impedance faults

Most categories are close to ceiling for all relays (Fig. 3). The exception is the high-impedance internal fault ($n = 459$ traces): 65.4 % for the deployed instance, 62.1 % for the tuned harmonic 87T and 51.0 % for the textbook settings; on the other internal faults the two are 96.8 % and 95.3 %. The instances detect 65.4–76.7 % of them, so a large share is missed by every relay, consistent with the known blind spot of percentage differential protection for low-level faults [36, 37, 38]. Switch-on-to-fault and external faults during inrush are the next weakest categories.

6.6. Operate time

The deployed instance trips with a median of 16.2 ms (90th percentile 36.2 ms), the textbook harmonic 87T with 16.9 ms (20.0 ms) and the unrestrained 87T with 1.2 ms (Table 5, Fig. 4). The 12.5 ms dwell sets a floor for every instance. Key's network is slower inside the logic (median 30.0 ms) than in its native mode (18.1 ms). The deployed instance is slower than the harmonic relay on faults during inrush (median 28.1 ms against 17.2 ms) and

Table 5: Operate time (ms) per method and fault class: median / 90th percentile over correctly tripped fault traces, plant-A CV pooled over 3 seeds × 5 folds, common set.

Group	Method	All faults	Solid LV / HV	CT saturation	In inrush	SOTF	High impedance	n (all)
Reproduction (native)	Simões	0.4 / 1.3	0.3 / 0.9	0.5 / 0.6	0.3 / 0.6	0.4 / 1.3	0.4 / 7.0	2,590
	Ngaopitakkul	5.1 / 65.2	5.1 / 51.4	–	5.2 / 674.1	5.1 / 5.3	5.0 / 5.3	1,030
	Abdusalam	21.1 / 28.5	21.1 / 27.9	23.7 / 40.6	18.7 / 21.2	21.1 / 27.7	19.0 / 21.5	2,839
	Afrasiabi	11.2 / 17.7	11.0 / 17.9	11.1 / 13.8	8.3 / 12.2	11.6 / 21.7	16.1 / 24.1	3,223
	Key	18.1 / 19.7	18.1 / 19.4	18.4 / 20.0	15.9 / 19.7	19.1 / 47.5	18.1 / 21.2	3,605
	Alhamd	15.2 / 20.2	15.0 / 20.0	16.7 / 20.6	14.2 / 19.5	19.4 / 21.9	12.8 / 19.8	3,588
Conventional relay	87T unrestrained	1.2 / 7.5	1.2 / 8.1	1.2 / 3.8	–	1.9 / 6.2	6.2 / 14.4	2,985
	Harmonic 87T, textbook (single pass)	16.9 / 20.0	16.2 / 19.4	17.5 / 20.0	17.2 / 377.8	16.9 / 87.3	17.2 / 18.9	1,065
Dual-path instance	LSTM, deployed	16.2 / 36.2	15.6 / 31.2	16.2 / 27.5	28.1 / 45.6	21.2 / 163.4	25.0 / 45.7	3,455
	Key's network	30.0 / 33.8	30.0 / 31.9	30.6 / 31.9	30.0 / 40.6	31.2 / 98.9	30.0 / 41.2	3,498
	1D-CNN	16.2 / 31.9	15.0 / 26.0	16.2 / 20.0	23.1 / 38.1	19.4 / 151.0	19.4 / 39.9	3,559
	GRU	15.0 / 29.4	14.4 / 23.1	15.0 / 21.2	20.6 / 38.8	18.4 / 46.6	19.4 / 43.0	3,553
	MLP	16.2 / 38.8	15.6 / 33.1	16.2 / 19.4	35.0 / 43.1	35.6 / 160.5	20.6 / 51.2	3,528
	Random forest	17.5 / 50.0	15.6 / 46.2	19.4 / 35.5	37.5 / 48.9	30.0 / 143.5	31.9 / 68.7	3,475
	LSTM, tuned	15.6 / 30.0	15.0 / 19.4	15.6 / 20.0	21.9 / 41.2	16.9 / 45.8	16.2 / 39.4	3,479
	SVM	17.5 / 37.5	16.2 / 35.0	16.9 / 35.6	35.6 / 47.4	35.6 / 40.4	31.9 / 38.1	3,234
Ablation	LSTM, deployed, gated only	16.2 / 33.1	15.6 / 26.9	16.2 / 27.5	28.1 / 45.6	21.2 / 163.4	20.6 / 36.2	3,287
	LSTM alone	15.6 / 31.9	15.0 / 22.5	15.6 / 20.0	23.8 / 41.4	18.8 / 112.9	18.1 / 39.4	3,630

Notes. Operate time = trip time minus the applied fault time (verified on the waveform by the generator check); counts are (seed, fold, trace) decisions, so each trace appears up to three times; descriptive, no test. Reproductions run at their own sampling rates and decide once per detector firing (Simões, Ngaopitakkul) or per window; the harmonic 87T row is a single pass over all traces with textbook settings (no operate times were saved for the tuned settings). Evolving faults are counted only in 'All faults'. '–' = no correctly tripped trace in the class.

on high-impedance faults (25.0 ms against 17.2 ms). Operate times are reported as measured from the applied fault time; no limit from a standard is claimed [39].

6.7. Shifted range and transfer to plant B

On the shifted-range set the deployed instance reaches 98.0 % / 96.0 % and Key's network 98.0 % / 95.3 % (single scoring, Table 3). On plant B the two diverge: the deployed instance keeps 92.7 % dependability but its security drops to 88.4 % (89 false trips among 764 healthy traces), while Key's network reaches 95.5 % / 99.9 % (1 false trip among 757). 87 of the 89 false trips of the deployed instance came first through the classifier-alone path, 62 of them on external faults with CT saturation (Fig. 5). With that path off, the same traces give 90.5 % / 99.2 %. The textbook harmonic 87T reaches 94.1 % / 99.0 % on plant B. Transfer is thus a property of the classifier, and the classifier-alone path is where a poorly transferring classifier does harm.

6.8. Closed loop and execution time

In the Simulink closed loop, the relay with the deployed instance agreed with the offline mirror on 50 of 50 traces (maximum trip-time difference 0.30 ms over 24 common trips) and was correct on 49; 48 of 50 cases met

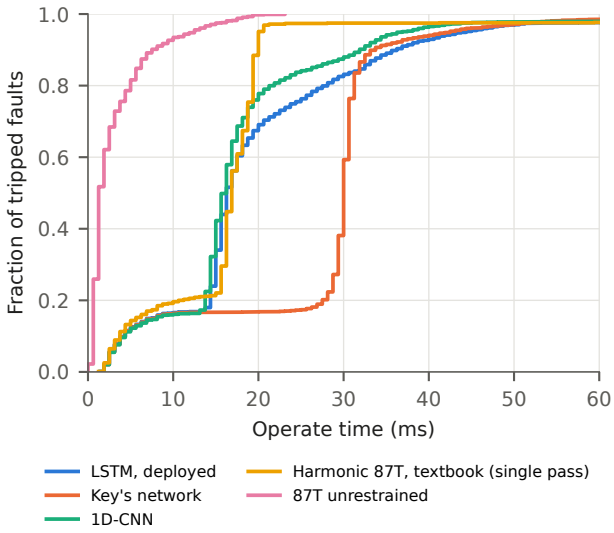


Figure 4: Empirical distribution of the operate time of correctly tripped faults (plant-A CV, common set).

the combined decision and 30 ms check, and the plant-B suite passed 12 of 12. Feature extraction and inference take a median of 139.6 μs with ONNX Runtime and 175.9 μs in ANSI C (99th percentile 228.9 μs) against a 625 μs sample period on a development PC; these are not relay-processor measurements. Among the reproductions, the feature stage of Abdusalam et al. needs 609 ms per window against a 2.5 ms budget, and Alhamd's trained network 1,166 μs against 1,667 μs.

7. Discussion

What "model-agnostic" means here. The supervisory layer (dwell, validation-tuned threshold, high-set, optional 87T gate) gives a similar operating point with 6 of 7 classifiers. The 87T gate itself is not what makes a classifier secure. For the poorly calibrated deployed LSTM, the gate raises security from 95.6 % (classifier alone, same dwell and threshold rule) to 99.0 % at a cost of dependability (97.7 % to 93.0 %). For Key's network, which is well calibrated, the like-for-like mode without the gate already reaches 97.2 % / 99.6 %, and the gate moves it to 94.1 % / 99.9 %: 3.1 points of dependability for 0.3 points of security. The gate is therefore a guard against a classifier

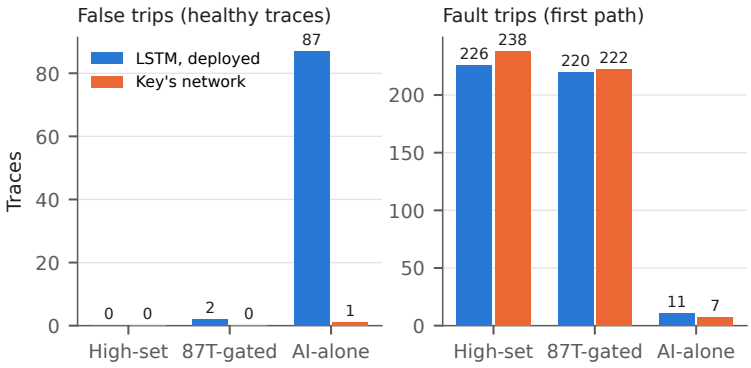


Figure 5: Plant-B zero-shot transfer: the path that tripped first, for false trips (left) and fault trips (right), deployed LSTM instance and Key's network inside the same logic.

one cannot trust, not an improvement of a good one.

The classifier-alone path is an option, not a default. It adds 4.5 points of dependability on plant A at a cost of 0.5 points of security, mainly on faults below the 87T pickup. On the unseen plant it caused almost all false trips of the deployed instance, and the plateau rule already switched it off in 3 of 15 CV fold-models (AI threshold 0.75–0.999). A utility applying the logic to a transformer it was not trained on should run the gated-only variant until the classifier has been retrained or validated on that transformer; with a classifier that transfers (Key's network) the path did little harm.

The conventional baseline. The tuned harmonic 87T is a stronger baseline than textbook settings, and the deployed instance's dependability lead over it is small and varies across seeds. Its grid optimum lies at the grid boundary, and cross-blocking or waveform-based restraint were not evaluated, so the lead of the learned instances over a fully optimised numerical relay may be smaller than reported here.

Benchmark lessons. Native and like-for-like scores of the same classifier differ by tens of points for some reproductions, so a comparison of published accuracies says little about the classifiers. The neural reproductions of Key and Afrasiabi were not bit-reproducible: their first and second runs on identical folds differ on 74 and 367 of 9,906 decisions; the tables use the second run.

8. Limitations

The study is a simulation of two lumped transformer models with terminal faults; there is no hardware-in-the-loop or field test, no turn-to-turn fault, no winding capacitance, no CT remanence [40] and no sympathetic inrush [41, 42]. Plant B fails 2 of its 6 physics criteria. No relay detects more than 76.7 % of the high-impedance internal faults. Several analyses are post-hoc (Section 5); the B2 search is not nested and gives SVM and random forest fewer training windows. The tuned harmonic 87T and the B2 instances were evaluated in CV only. Execution times come from a development PC. The reproductions are re-implementations trained on our data, not the authors' code, and those of Simões et al. and Ngaopitakkul and Kunakorn run off their design point on these plants.

9. Conclusions

1. On a leak-free benchmark, the decision rule of a learned 87T scheme matters as much as its classifier: 0 of six reproduced methods meets 93 % dependability and 98.8 % security in its native mode, and 1 in the like-for-like mode.
2. A dual-path supervisory logic reaches a similar operating point with 6 of 7 classifiers, and each of them is more dependable than a validation-tuned harmonic-restrained 87T by 2.3–4.6 points at comparable security.
3. Transfer to an unseen transformer depends on the classifier, and the classifier-alone path is the weak point under transfer; the gated-only variant is the safe choice for an untrained plant.
4. High-impedance internal faults remain the limit of all relays tested.

The benchmark (folds, per-trace decisions, specifications and generators) is public, so further learned 87T schemes can be compared on the same terms.

Data and code availability

The Simulink plant models, data generators, evaluation code, reproduction specifications, per-trace decisions of every method and the scripts that build every table and figure are available at `https://github.com/SS-Abeer13/Hybrid-DWT-LSTM-Transformer-Relay` ([**AUTHOR:** release tag and commit]). The simulated datasets and trained models are [**AUTHOR:**

deposited at a DOI, or available on request because of their size; they can be regenerated with the listed scripts in MATLAB/Simulink release and toolboxes to be stated]. All statistics and tables can be recomputed from the per-trace result files without MATLAB. No field data were used.

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[**AUTHOR:** competing-interest statement]

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[**AUTHOR:** the author's own declaration: tool, purpose, and that the authors reviewed and edited the output and take full responsibility (Elsevier policy). Research use of the AI coding assistant is described in Section 5.]

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